MADHAV GARG

Third-year B.Tech Computer Science and Engineering Student

JK Lakshmipat University, Jaipur, Rajasthan, India | Location: Jaipur, India

Email: madhavgarg.agra@gmail.com | **LinkedIn:** linkedin.com/in/madhav-garg-agra | **GitHub:** github.com/madhavgargagra

Computer Science and Engineering student at JK Lakshmipat University, Jaipur, interested in full-stack development, AI, software engineering and emerging technologies. Experienced in building practical applications, exploring new tools, solving problems and collaborating with teams. Developed leadership and coordination experience through university projects, internships and cultural initiatives.

EDUCATION

JK Lakshmipat University (JKLU) — 2024 — 2028 (Expected)

Bachelor of Technology in Computer Science and Engineering

Current Status: 3rd Year (Semester 5) | Cumulative GPA (CGPA): 8.6 / 10.0 | Jaipur, Rajasthan, India

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Full Stack Web Development, Operating Systems, Machine Learning Foundations, Computer Networks, Software Engineering

TECHNICAL SKILLS

- **Languages & Core:** JavaScript (ES6+), Python, HTML5, CSS3, SQL, C/C++ fundamentals
- **Frontend Architecture:** React 19, React Router, Vite, Tailwind CSS v3, Framer Motion, Responsive UI/UX, Web Accessibility (A11y)
- **Backend & APIs:** Node.js, Express.js REST APIs, FastAPI (Python), JWT Authentication, CORS Security
- **Databases & ORMs:** MongoDB (Mongoose ODM), MySQL (Sequelize ORM), SQLite
- **Machine Learning & AI:** Scikit-learn (Isolation Forest, SVM, KNN, Logistic Regression), XGBoost, TensorFlow.js, MobileNet v2
- **Geospatial Analytics:** Google Earth Engine, QGIS, Sentinel-5P, MODIS, Landsat-8, Urban Heat Island (UHI) & Aerosol Analysis
- **Developer Tools & Systems:** Git, GitHub Actions CI, Docker containerization, Pytest, Postman, Linux CLI / Red Hat RH124

PROJECT PORTFOLIO

EcoSort — Waste Sorting & Environmental Learning Platform — Open Source & Live Demo

Stack: JavaScript, HTML5, CSS3, TensorFlow.js, MobileNet v2, PWA

- **Browser AI Inference:** Delivered client-side waste classification using TensorFlow.js and MobileNet v2 with camera/image upload and voice search.
- **Educational Features & PWA:** Incorporated sorting game, recycling-myth quiz, carbon impact dashboard, and offline functionality via Service Workers.

[GitHub Repository] **[Live Demo]**

Lung Cancer Prediction & Web Application — Machine Learning Evaluation

Stack: Python, Flask, XGBoost, Scikit-learn (SVM, KNN, Logistic Regression)

- **Algorithm Comparison:** Trained and evaluated four supervised models on clinical indicator datasets with model explanation and report generation.
- **Educational Scope:** Educational machine-learning project exploring predictive models; not clinically validated or intended for medical diagnosis.

[GitHub Repository]

AI-Powered Insider Threat Detection System — Enterprise Security Analytics

Stack: FastAPI, Next.js, React, SQLite, Scikit-learn (Isolation Forest), Docker, Pytest

- **Security Analytics Architecture:** Engineered multi-role enterprise security system evaluating 5-factor weighted risk models for anomalous employee activities.
- **Private Internship Project:** Developed during Infosys Springboard internship program; access restricted to maintain enterprise confidentiality.

Festival Event Management System — Full-Stack Relational Database

Stack: React, Express.js REST API, MySQL, Sequelize ORM

- **Relational Database Architecture:** Engineered full-stack festival event management application backed by MySQL and Sequelize ORM modeling relational schemas and role-aware administrative workflows.

[GitHub Repository]

INTERNSHIP EXPERIENCE

Infosys Springboard
July 2026 — October 2026

Full-Stack Developer Intern | Project: AI-Powered Insider Threat Detection System | Remote

- **Enterprise Backend:** Architected decoupled FastAPI backend with role-based JWT authentication for 4 distinct SOC roles and SQLite database.
- **Machine Learning Anomaly Detection:** Integrated Scikit-learn Isolation Forest unsupervised model with weighted five-factor risk scoring for employee behavior analytics.
- **Testing & DevOps:** Implemented automated Pytest unit/integration test suites, Docker containerization, and GitHub CI workflow pipelines.
- **Project Scope & Notes:** Project notes report simulated alert triage reduction from ~90 to 23 actionable incidents (project-reported metric; not an independently validated production deployment). Certificate awaited; repository is strictly private.

India Space Academy
19 May 2026 — 2 July 2026

Remote Sensing & GIS Intern | Summer Training Program (6-Week Completed) | Hybrid

- **Urban Heat Island Analysis:** Processed multi-spectral satellite datasets to map Land Surface Temperature (LST) and Built-up Density indexes across regional corridors.
- **Air Quality & Aerosol Mapping:** Utilized Sentinel-5P and MODIS satellite data in Google Earth Engine and QGIS to analyze atmospheric aerosol patterns across the Taj Trapezium Zone (TTZ) and Delhi NCR.
- **Evaluation & Defense:** Maintained official training logbooks, authored technical research documentation, and successfully defended findings during technical presentation and viva.

LEADERSHIP, CULTURAL & CAMPUS ACCOLADES

- **Chair, Student Drama Club (2025 — 2026):** Led an active student theatrical society of 25+ members across scriptwriting, ensemble rehearsals, and campus stage productions.
- **Cultural Coordinator & Decor Volunteer:** Contributed to venue decor, backstage logistics, and audio-visual preparation for Sabrang, JKLU's annual cultural festival.
- **Athletic & Cultural Honors:** Runner-up in University Football Championship; Second Position in Nukkad Natak (Street Play) competition at Sabrang Fest.
- **Campus Representation:** Represented JKLU in inter-college drama competition at University Maharani College; participant in HackJKLU hackathon, Muqabala & Spardha.

SELECTED CERTIFICATIONS & LICENSURES

- **Red Hat System Administration I (RH124 - RHA):** Certificate of Attendance (15 April 2026, 40 credit hours). Covers Linux CLI, user management, and process control.
  [View Official Credly Badge]
- **Infosys Springboard Certified:** Artificial Intelligence Primer (1 May 2026), Deep Learning for Developers (1 May 2026), Generative AI Unleashing (1 May 2026), Introduction to Data Science (1 May 2026), Agile Scrum in Practice (1 May 2026).
- **Additional Coursework:** Database Management Systems (DBMS Part 1 & Part 2), Computer Vision 101, Introduction to OpenAI GPT Models, Prompt Engineering.